

YUMI MATSUYAMA

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SUMMARY

Bilingual Ph.D. Candidate with a mechanical engineering degree, offering 8 years of laboratory experience in combustion and biomedical research, 9 years of teaching/tutoring in various disciplines, CAD modeling, MATLAB programming, and R&D experience.

EDUCATION

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| 2019-Present | Doctor of Philosophy in Mechanical Engineering, Fluid and Thermal Sciences
Cleveland, <i>Case Western Reserve University</i> (expected grad. Sept 2021) |
| 2016-2018 | Master of Science in Mechanical Engineering
Cleveland, <i>Case Western Reserve University</i> |
| 2012-2016 | Bachelor of Science in Mechanical Engineering, Minor in Japanese
Cleveland, <i>Case Western Reserve University</i> |

RESEARCH EXPERIENCE

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| 2016-Present | Research Assistant, <i>Case Western Reserve University</i> , Cleveland, Ohio
Dept. of Mechanical and Aerospace Engineering
• Research Advisor: Dr. Fumiaki Takahashi <ul style="list-style-type: none">○ Designed experiments to investigate toxic gas detection and analysis with Smoke Density Chamber and Fourier Transform Infrared Spectroscopy gas analyzer, and model with Fire Dynamic Simulator○ Observed how moisture content, orientation, density of wood species affects smoke toxicity○ Working with FEMA and USDA Forest Service to validate hand-held device prototype for irritant and particulate detection to aid fire responders○ Maintains Cone Calorimeter and Microscale Combustion Calorimeter |
| 2012-2016 | Lab volunteer, <i>Case Western Reserve University</i> , Cleveland, Ohio
Dept. of Mechanical and Aerospace Engineering, 2014-2016
• Research Advisor: Dr. Umut Gurkan <ul style="list-style-type: none">○ Developed MATLAB program for red blood cell deformation analysis Dept. of Ophthalmology and Visual Science Laboratory, 2012
• Research Advisor: Dr. Tadao Maeda <ul style="list-style-type: none">○ Researched effects on visual cycle with selected protein-receptor deficient mice Dept. of Hematology and Oncology, 2010-2012
• Research Advisor: Dr. Shigemi Matsuyama <ul style="list-style-type: none">○ Wet-lab environment – performed PCR via gel electrophoresis, tasked with pipetting and agarose gel formation, mixed paraformaldehyde solutions for organ preservation, lab mice care |

- 2014 Intern, *Innovative Delta Technology*, Moreland Hills, Ohio
- Worked under Michael Nilsson, PE
 - Designed surgical instruments and implants for spinal surgery using SolidWorks
 - R&D and aided in patenting equipment
 - Consulted and translated for clients from Japan during conferences
 - Reviewed and translated patents from Japanese to English

TEACHING EXPERIENCE

- 2012-Current Teaching Assistant, *Case Western Reserve University*, Cleveland, Ohio
- EMAE (Mechanical and Aerospace Engineering Department)
 - 353: Heat Transfer (1 semester)
 - 252: Fluid Mechanics (2 semesters)
 - 285: Mechanical Measurements Engineering Laboratory (3 semesters)
 - ENGR (General Engineering Department)
 - 210: Introduction to Circuits for Engineers (2 semesters)
 - JAPN (Japanese Language Department) - tutor
 - 101 – 351: All undergraduate levels for Japanese Language (4 semesters)
- 2018-2019 Mentor for four CWRU undergraduate students for their senior capstone

HONORS AND AWARDS

- 2018 Graduate Student Travel Award, Case Western Reserve University
Student Presenter Travel Award, University of Illinois at Urbana-Champaign
- 2012-2016 Honorary University Scholarship receive for all of undergraduate career
- 2014 Japanese Book Award – highest GPA in Japanese Language Dept. of 2013-2014
- 2013 CWRU Sports Club of the year (during presidency for Case Kendo Club)

PUBLICATIONS

- 2019 . **Y. Matsuyama**, “Toxic Gas and Particulates Characterization in a Smoke Density Chamber,” M.S. Thesis, Mechanical and Aerospace Engineering Department from Case School of Engineering, Case Western Reserve University, Cleveland, Ohio, Published 2019. OhioLink.
- 2016 . Y. Alapan, **Y. Matsuyama**, Little J. A., Gurkan U. A. “Dynamic Deformability of Sick Red Blood Cells in Microphysiological Flow.” *Technology*. [Online]. Vol 4. (2016, Feb.).
- Manuscripts in preparation . **Y. Matsuyama**, F. Takahashi, “Effects of Surface Orientation on the Smoke Toxicity of Flaming and Pyrolyzing Wood”
 . **Y. Matsuyama**, F. Takahashi, “Relative Toxicity of Smoke from Burning and Pyrolyzing Polymers”

PRESENTATIONS

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| 2020 | . 13 th International Association of Fire Safety Science in Waterloo, Canada
(moved to 2021 remotely) |
| 2018 | . 37 th International Symposium on Combustion in Dublin, Ireland
. 2018 Central States Section of The Combustion Institute in Minneapolis,
Minnesota |
| 2015 | . 2015 Midwest Regional Meeting of the American Society of Biomechanics in
Akron, Ohio
. Case Western Reserve University Research ShowCASE |